



Detecting Phishing Emails by Identifying Common Phishing Characteristics.

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Abstract:-

Phishing mail: Phishing mails the most common cybersecurity threat followed by malwares, but phishing is one of those attacks where you will never notice anything wrong. This project aims to identify phishing emails that contain certain traits of phishing, including suspicious links, urgency language, false sender address, grammatical errors and misleading attachments. This paper analyzes these attributes using machine learning and text analysis techniques to categorize emails as phishing or legitimate. This involves analyzing and preprocessing either a dataset with both phishing and genuine emails or even just based on one of the dataset to train classification models. This helps Identifying similar phishing patterns, which improves e-mail safety and ensures the attacks on cybersecurity are highly reduced. This study demonstrates the potential impact of automated phishing detection systems in safeguarding users and organizations from continually changing threats in the online landscape.

Introduction:-

Phishing e-mails have emerged as one of the most popular forms of cyber security attack, with the purpose of tricking victims into disclosing sensitive data such as passwords, bank account details, and personal data. Attackers use a number of tricks to deceive users using phishing e-mails by making them appear as being legitimate and sending them to trick victims with misleading URLs, spoofing e-mail addresses, attaching malicious files, and writing deceptive messages. However, with the emergence of sophisticated phishing techniques, conventional filters cannot adequately detect phishing attempts as legitimate e-mails can be mistakenly identified as phishing. Some of the earlier approaches included blacklisting software, browser plug-ins, e-mail verification services, and password-based filters. Although these methods offered some form of protection against phishing, a considerable drawback in several instances was a high rate of false positives, frequent manual updates, limited examination of e-mail contents, or failure to adequately protect against new attacks. It is for this reason that automated phishing detection systems based on machine learning algorithms and data mining techniques have been widely studied due to their capacity to examine e-mail contents and determine suspicious aspects automatically. Studies indicated that several types of classifiers such as logistic regression, decision tree, support vector machines, random forests, neural networks, and fuzzy systems can achieve good results if the right set of email features are identified and used. Such features often include aspects from the e-mail body, header data, URLs, JavaScript behavior, and other external aspects. Thus, the present project will aim to detect phishing e-mails using their common features and classification algorithms. [32]



Literature Review:-

I. Phishing Fundamentals & Types of Attacks:

i. Definition of Phishing Attacks

Phishing is a social engineering-based cyberattack, where the attacker tricks victims into revealing some sensitive information, such as usernames or passwords or even financial information. The attackers in such attacks usually pose as trusted parties to gain the victim's trust, such as banks, organizations, or online services. In contrast to the traditional cyberattacks that primarily use technical flaws, phishing attacks are more focused on human behavior, which makes it highly effective and commonly used [1]. Moreover, studies have shown that phishing attacks are typically organized. The first step involves the attacker crafting a fraudulent but legitimate-looking message and will then transmit it either through email, messaging systems or malicious websites. The target is then persuaded to click on the content, usually a link or an attachment. Finally, the victim is tricked leading to system compromise or data theft. This is commonly referred to in the literature as a lifecycle that involves stages like luring, interaction, and exploitation [1].

ii. Evolution of Phishing Techniques

Phishing attacks have evolved over time and have become more advanced and focused. According to the previous research, phishing attacks in the 1990s were relatively simple and primarily relied on email-based scam messages that imitated a financial institution or a well-known online service. Over time, these techniques were refined, particularly in the 2000s, as social engineering was used more. Attackers began personalizing their attacks to make them more credible, in the form of spear phishing, in which specific individuals or companies are targeted. Further, according to studies, in the 2010s, phishing attacks were more advanced with new techniques such as whaling targeting high-profile individuals, smishing or vishing based on text messages or phone calls. More recently, phishing has also evolved into sophisticated multi-vector attacks where there is a combination of different channels, such as emails, social media, and instant messaging. In addition, researchers indicate that cybercriminals are adopting more sophisticated technologies like artificial intelligence to develop more authentic and automated phishing attacks, which are harder to detect [2].

iii. Types of Phishing Attacks

There are various types of phishing attacks identified on the basis of their delivery methods and targeting approaches. Studies show that the most common type is email phishing where attackers send the same message to a large number of users. However, spear phishing is more targeted, as attackers adjust messages for specific people or groups to make it more believable. In a similar way, whaling focuses on high-profile individuals, such as CEOs and politicians. The literature additionally describes another form of phishing, which is clone phishing where attackers mimic legitimate emails and alter their content to contain harmful links or attachments. These categories show how phishing techniques have evolved to integrate both technical and social engineering methods to enhance the effectiveness of these attacks [3].



iv. Email Spoofing Techniques

One of the most common techniques in phishing attacks is email spoofing where attackers forge sender identities and masquerade as trusted parties to trick their victims. Studies indicate that spoofing is normally implemented by the manipulation of email headers, including header fields such as the “From” and “Return-Path” addresses, which enable the attackers to conceal the actual source of the message. Moreover, the research indicates that the method takes advantage of the vulnerabilities existing in email protocols, allowing malicious actors to transmit fake messages that seem to be authentic. Moreover, email spoofing usually accompanies social engineering techniques to enhance the success of phishing attacks, making it more difficult to distinguish between authentic and malicious messages [6].

v. URL Obfuscation Methods

One of the most widely used techniques in phishing attacks is URL obfuscation where the attackers manipulate URL structures to hide their actual destination and make them appear legitimate. Studies have shown that it does this by changing various elements of the URL thereby making it difficult for users to recognize the bad links. Common techniques include the replacement of domain names with IP addresses, URL shortening systems, port numbers, and encoding schemes like Punycode to make domains visually misleading. Moreover, research highlights the fact that these methods are not only aimed at bypassing technical detection systems but also at exploiting user perception to enhance the success of attacks. These results show that URL obfuscation is a critical component of phishing, as it is used to blend technical and user-deception techniques [7].

I. Common Characteristics of Phishing Emails:

i. Common Phishing Indicators

Phishing emails can be analyzed based on multiple structural features, such as email headers, email bodies, and embedded URLs. Studies indicate that the header information, including sender details and routing details, can indicate spoofing attempts, whereas content-based features involve the use of deceptive language and misleading messages meant to be perceived as valid. Moreover, URL-based features, including suspicious or improperly matched links, are usually employed to redirect victims to malicious websites. Such a combination of attributes makes phishing emails difficult to detect because attackers carefully design both the technical structure and content of the message to appear like a legitimate correspondence. Moreover, it has been highlighted in the literature that a combined analysis of these characteristics is a better way to detect phishing emails than a single indicator [5].



ii. Role of Urgency and Fear Tactics

When looking at psychological tactics, attackers often apply perceived urgency as the most prevalent technique by pushing their targets to act on a phishing email before any rational thought kicks in [8]. Attackers create an “act now” scenario, which discourages the recipient’s own thinking or checking things by using direct expressions like “immediately” and “soon.” This pressure, in return, breeds intuition-driven responses and heightened tendencies to yield with a reduced probability of being noticed.

The article [8] also emphasizes the role of fear-based persuasion, where attackers threaten negative consequences such as account suspension, data loss, or financial harm if no action is taken. Urgency and fear are often used together, with 82.54% of phishing emails marked as urgent and 71.42% having both traits. The study also shows that removing these features leads to the largest drop in detection performance, confirming their importance as strong indicators. Overall, “act now” strategies are effective because they exploit human psychology, making urgency and fear essential features for both user awareness and automated phishing detection systems.

iii. Linguistic Features of Phishing Emails

It is these linguistic features that are used to identify phishing emails, and as such, they encompass grammar and tone, as well as writing style [9]. Messages tend to have spelling mistakes and grammatical errors (although these are not as common). For a variety of reasons, attackers use capital letters, exclamation marks, or even unusual symbols to draw attention and make their emails appear more urgent. Phishing subject lines like this one: “URGENT!!! Your acc0unt has been suspended – act now!!” demonstrate how errors and exaggerated punctuation can be used to manipulate users. However, the study also indicates that more sophisticated phishing emails minimize obvious mistakes to appear more legitimate, making grammar errors a weaker standalone indicator.

Furthermore, the study in [9] explains that tone and writing style differences are crucial linguistic indicators. Interestingly enough, 58% of phishing emails are neutral and 35% positive in sentiment — but they are never aggressive or threatening. This helps attackers to impersonate a real organizational communication without any trace. For instance, examples such as “Important Security Message From HSBC Bank UK” and “Nationwide Account Notification” appear professional and trustworthy despite being malicious. Also, phishing messages are concise (4–6 words), to the point, and targeted with pronouns like “your”, such as “Your account has been limited”, to make it relevant. And the style emphasizes formal vocabulary, with nouns like “account” and “security” and “update,” making it feel authoritative. These patterns illustrate how linguistic markers used by attackers are chosen to strike a delicate balance between perceived authenticity and persuasive force in order to frustrate conventional phishing detection techniques.



II. Human Factors & Social Engineering:

i. Social Engineering in Phishing

Social engineering is a key aspect of phishing attacks since the latter is mainly based on manipulating human behavior instead of technical weaknesses. It has been shown that deception, persuasion, and urgency are some of the psychological tactics employed by attackers to manipulate victims and enhance the chances of responding. As an example, elements like email subject line, message format, and perceived legitimacy of a sender have been demonstrated to play a great role in influencing the user's vulnerability to phishing attempts. In addition, studies indicate that the individuals' habits and attention to email cues might be a determinant of their vulnerability, since users who fail to carefully evaluate the features of the messages are more likely to fall victim to such attacks. All in all, the literature has shown that phishing is not just a technical menace but a human-centered attack that capitalizes on cognitive and behavioral vulnerabilities [4].

ii. User Awareness and Training: Human factor in preventing phishing

The study [22] looks at simulation-based training tools for common phishing threats and prevention. It shows that phishing is still one of the main causes of data breaches, while better user awareness can help combat these threats. This study presents a simulation system that can simulate real phishing techniques from different channels, like email, SMS as well as voice-based attacks. The results show a profoundly increased defence against user susceptibility in the case of combining regular training with simulated attacks, enabling quantification of success at leading people to better engage in desired behaviours such as link and sender identity verification. This piece has some strong points, one of which is its practical approach to improving human-centered security through continuous training. However, the study focuses more on user behavior and awareness rather than technical detection methods, which leaves a gap in automated phishing detection solutions. This limitation is important for projects that need to combine intelligent detection systems with user training.

III. Technical Detection Approaches:

i. Rule-Based Detection Approaches

Detection Approach: Rule-Based Detection approaches can be categorized as one of the earliest and most interpretable methods in phishing email detection, according to an academic research paper [11]. These strategies depend on an organized arrangement of a few principles or heuristics that wake up suspicious designs in email components like subject line, body content, sender address, and embedded URLs. Rule-based systems, unlike machine learning-based techniques, do not require large training datasets or complex computational resources and therefore work well even for low-resource environments. The findings indicate that rule-based classification is a viable approach for identifying phishing attacks on sites by scanning against known properties of phishing with low computational cost. Typically, these systems function by matching email features against a list of predefined indicators, such as the presence of urgent keywords ("verify," "urgent"), suspicious domains, or abnormal formatting patterns. When such features are detected, the system assigns scores or flags, enabling classification of emails as phishing or legitimate.



A common implementation of rule-based detection involves a scoring mechanism where multiple rule matches increase the likelihood of an email being classified as phishing. As described in the provided study, rules may include keyword detection, URL structure analysis (e.g., misspelled domains like “g00gle.com”), detection of malicious file extensions, and identification of unusual encoding patterns. Supporting research demonstrates the effectiveness of rule-based systems to identify phishing, exploiting URL and structural features, because attackers are more vulnerable to predictable deceptiveness, such as subdomain abuse and deceptive links. Also, heuristic-based approaches enhance rule-based detection by adding domain knowledge of phishing behaviors to existing systems, improving detection quality without introducing a further layer of complexity. While rule-based systems have their pros and cons, they also suffer from limitations: They do relatively poorly when it comes to identifying new and/or highly advanced phishing attacks that would not match current rules. However, this is offset by their transparency, ease of implementation, and low computational cost—making them a useful part of multi-layered email security systems. [11]

ii. Blacklist and Whitelist Techniques

Blacklists are among the earliest and most common methods for filtering phishing emails and URLs. Blacklist-based systems check incoming URLs or domains against a database of known malicious sources. This makes them quick and easy to use. However, they rely on prior knowledge. Any phishing site that hasn't been flagged can slip through undetected. Whitelist-based approaches work differently, as they only allow access to trusted domains. This can help limit exposure to unknown threats. Azeez et al. proposed an automated whitelist method that cross-checks visual links, actual hyperlinks, trusted domains, and IP addresses to improve phishing detection. Even with this improvement, both methods have limitations. Attackers often register new domains and slightly change URLs to avoid detection by existing lists [12].

iii. Machine Learning for Phishing Detection

Machine learning steps in because it learns from examples of strict rules. Classifiers like Bayes or Support Vector Machines look at keywords that seem off sender details, URLs and how the email is built. Random Forest and those others do things to sort out the fake emails from the real emails. Murti and Naveen tried it out. Naive Bayes is okay for text but Support Vector Machines do better at pulling apart phishing emails from legitimate emails when the right features are picked. The catch is, if your data is old or not labeled well the whole thing falls apart when it comes to accuracy. It depends a lot on what you feed the machine learning[13].

iv. Deep Learning Approaches

Deep learning takes it further with networks that pick up on tricky patterns in the email text, URLs and extras without you having to hand pick every feature. Things like Convolutional Neural Networks, LSTMs or even BERT models get used for this. They catch the deeper



language connections that regular machine learning might skip over. Thakur and team reviewed a bunch. Said deep learning is getting big, in phishing work especially for messy datasets if you tune them right. Deep learning boosts detection. You need tons of data and heavy computing power. Training has to be careful so the deep learning does not overfit and fail on new emails. The part that stands out is how resource heavy deep learning can be[14].

v. Natural Language Processing (NLP)

Machine learning is useful because it learns from examples. There are classifiers like Bayes and Support Vector Machines. These classifiers look at keywords that seem unusual. They also look at the sender. It helps detection systems understand what the email says and how it says it. Natural language processing techniques can find patterns in the text. These patterns include things that try to create a sense of urgency. They also include manipulation. However, phishing emails sometimes pretend to be from trusted brands. The writing style may not match what a legitimate organization would send. Research by Salloum et al. shows that combining natural language processing with machine learning is an idea. This combination can improve phishing detection. It does this by looking at the email body. It pulls out features that are not just about the structure of the email. This matters because some phishing emails look clean. The language used is still trying to deceive. There are some limitations to this approach. Messages in languages can also be hard to flag. Content generated by intelligence tools can be tricky too. All of these can be harder to detect using analysis alone [16].

The use of the BERT transformer model in the classification of phishing email has been explored by Otieno et al. (2023). This study's goal was to achieve better phishing detection by using a deep learning transformer-based method that is able to interpret the semantics and context of email text. The researchers used phishing emails from the Nazario phishing corpus, legitimate emails from the Enron dataset and applied pre-processing, sentiment analysis, topic modeling, classification processes. The results revealed that the BERT model outperformed with an accuracy of 93%, and an F1 score of 93%, demonstrating that transformer-based models can enhance phishing email detection. The study also showed that traditional methods, such as sentiment analysis and feature correlation, are not always effective in classifying phishing emails because phishing attacks continue to change. Therefore, the researchers suggested using advanced deep learning methods such as BERT to improve phishing detection systems [26].

A recent study conducted in 2024 compared the traditional machine learning models and transformer-based models in phishing email detection. The study evaluated various traditional classifiers against state-of-the-art transformer-based classifiers like BERT, DistilBERT and RoBERTa. The analysis revealed that transformer-based methods outperformed the traditional machine learning approaches due to their ability to comprehend the semantics and context of phishing emails. In particular, RoBERTa achieved the best



detection performance with very high accuracy, demonstrating the effectiveness of transformer architectures in detecting complex phishing attacks. The study further revealed the drawbacks of the traditional machine learning models and the advantages of the transformer models in capturing the contextual information and adaptability of newer phishing detection systems, given the constantly changing nature of phishing language [27].

vi. Hybrid Deep Learning Frameworks for Phishing Detection

Alsubaei et al. (2024) have recently introduced a novel hybrid deep learning model for real-time phishing detection in cybercrime forensics. This framework incorporated several advanced techniques such as ResNeXt, Gated Recurrent Units (GRU), ensemble learning, feature engineering and SMOTE data balancing to improve phishing detection performance. The proposed model aimed to improve adaptability to the changing phishing attacks and to minimize false positive and false negative rates. The researchers tested the framework on a set of real-world phishing datasets and compared its results with some popular models like BERT, CNN, ResNet, and SVM. The results showed that the proposed hybrid approach had an accuracy of 98% which was higher than several state-of-the-art approaches, and fast execution time. The study also underscored the need to integrate various deep learning and feature extraction approaches to enhance the robustness, scalability, and real-time phishing detection capabilities of cybersecurity systems in modern cybersecurity environments [28].

IV. Feature Engineering & Data Analysis:

i. Feature Extraction Techniques

Phishing detection is a deal and it starts with something called feature extraction. This is where we take an email and turn it into something that our detection models can understand. We can break down the features into three groups. First there are text-based features. These are things like keywords that seem suspicious, phrases that try to create a sense of urgency, grammar that's not quite right and the overall tone of the email. Then we have URL-based features. This includes things like how the domain name is, whether the email uses an IP address instead of a domain name, how many subdomains it has, whether the brand name is misspelled and if there are any hidden redirects. Lastly we have metadata-based features. These look at the senders address, the email headers, the reply-to field and the authentication records. Phishing emails are tricky because they can put their signals in many different parts of the email. Research by Shajilal shows that using features from all three categories is better than using one. This is because phishing emails often spread their signals across the entire message rather than putting them all in one place [15].



ii. Dataset Usage in Phishing Detection

Public datasets are extensively employed in phishing detection research studies to aid in training and testing machine learning algorithms. For instance, datasets like Enron and SpamAssassin consist of examples of both valid and spam emails, enabling scientists to create better detection solutions. While such data can be applied for benchmarking and boosting classification performance, one of the major problems here is the lack of contemporary data that reflects current trends in cyber attacks; thus, a combination of datasets might be beneficial [17].

iii. Header Analysis in Email Security

According to the academic research paper [10], header analysis is an important approach for phishing email detection as it involves analyzing the metadata contained in an email instead of its actual content. An email is basically composed of two parts: the header and the body. The header includes such fields as From, To, Subject, and some routing-related information that offers a deep understanding of where the message has come from and how it is routed. While the body of an email is not that difficult to manipulate, to look legitimate, parts of header fields often give themselves away by being at odds with other parts -- hostile intent is all too clear. As an illustration, in the case of phishing emails, the sender name displayed may look legit, but the actual email address or domain is different from trusted sources (a.k.a Email Spoofing). Moreover, header fields like "Received" can show the order of mail through various servers, allowing analysts to (for example) follow a trail of abnormalities in domains, IP addresses, or improbable geographical origins.

Domain information, sender authenticity, and routing paths are important features that enable a system to specify the differences between legitimate emails and spam, helped by header-based features [10]. However, existing studies tend to underutilize these features compared to email body content, despite their high reliability in exposing attacks. In a similar fashion, differences between the sender domain they claim to be from and the sending server's actual domain, lack of proper authentication methods (SPF or DKIM), and irregularities with reply-to addresses are all good indicators of some phishing attempts. Similarly, while attackers may modify header data to impersonate trusted groups, a structured analysis might show obscure mismatches within the headers (for instance, contradictory sender identities within the mail loop or non-standard business takeovers). According to that, header analysis in conjunction with Machine Learning and feature extraction techniques makes the phishing detection systems more efficient because it provides an additional level of validation that is less vulnerable to spoofing.

V. Evaluation & Performance Metrics:

i. Evaluation Metrics

Evaluation metrics are an important part of assessing phishing detection algorithm performance. Some common metrics are accuracy, precision, recall, and the F1 score, and each one focuses on a different aspect of the problem. Accuracy shows overall classification results, precision is the percentage of predicted phishing emails that are actually phishing, recall shows how many phishing emails are correctly identified, and the F1 score balances



both precision and recall. The use of different metrics is necessary since the accuracy rate is insufficient to estimate system efficiency [18].

VI. Security Mechanisms & Protocols:

i. Email Security Protocols: SPF, DKIM, DMARC.

Research in [23] studies the adoption of artificial intelligence for adaptive phishing detection so as to overcome the caveats with existing solutions, namely blacklists or web heuristics based on machine learning techniques. The study presents a proposed system that uses machine learning algorithms and natural language processing in order to analyze the content of an email, metadata information, and URLs. For example, it was mentioned that the system can be trained on a combination of phishing and benign datasets made available so as to give it leeway into the new patterns of attack. Experimental results show that detection accuracy is supported by better true positive rates and lower false positives than traditional methods. The main advantage of this work is also one of the biggest pros — its flexibility to adapt to evolving phishing techniques. However, deploying these systems comes with issues regarding waveform scalability, data privacy, and preserving low error rates in real-world environments. These issues provide an important clue for members of any project intending to implement a phishing detection system; the AI will probably not be some sort of plug-and-play distance learning service.

VII. Challenges & Limitations:

i. Limitations of Existing Detection Systems

Despite numerous advances made within phishing detection, some challenges still remain relevant today. The first one refers to the fact that currently available systems cannot adequately adjust to emerging phishing techniques since they keep updating emails, websites, and senders' information to escape the detection process. Moreover, another limitation of such techniques is a high number of false positive results when the system identifies legitimate emails as phishing messages. Additionally, there are certain concerns related to the dependence of many machine learning solutions on the quality of the used database. It makes them rather vulnerable and prone to errors [19].

ii. Zero-Day Phishing Attacks

Zero-day phishing attacks represent those phishing schemes that were not previously encountered in databases or blacklists. Such attacks are challenging to detect due to the absence of any signs associated with previous attacks. In most cases, attackers use new domains, update email structure, and utilize other innovative approaches to make their messages look like regular ones. Rule-based approaches are often inefficient in dealing with zero-day attacks while anomaly-based and machine learning approaches prove to be more successful. However, identifying brand new patterns can still be challenging [20].



iii. Adversarial Attacks on Detection Models: Attackers bypassing ML systems

The application of artificial intelligence techniques to detect adversarial attacks in AI-based intrusion detection systems is examined in [21]. Detection networks specific to the financial pen, trained to be less vulnerable to inputs expressly tailored with the intent of bypassing classic systems. The study analyzes the mechanisms of adversarial samples to show that AI-based security systems are vulnerable and offers some ways to make them less so by examining how learning algorithms can be influenced. Focusing on building robustness of reliable detection in hostile environments is one of the strengths of this work and a necessity for modern cybersecurity applications. However, the study mainly focuses on adversarial intrusion scenarios instead of phishing-specific threats, which limits how directly it can be applied to email-based attack detection. This creates a gap for projects that focus on phishing attacks, where similar robustness principles must be adapted to different types of attacks.

Shirazi et al. recently conducted a study on adversarial sampling attacks on machine learning-based phishing detection systems. The researchers looked at the phishing attack surface of phishing sites and how attackers can exploit those features to fool well-trained machine learning classifiers without having to know the actual learning algorithm. The study showed that phishing detection mechanisms are extremely susceptible to adversarial learning techniques that are used by attackers to purposely change certain characteristics to bypass a detection system. Experimental results showed that changing just one feature had a significant impact on the phishing detection rate, and changing 4 features reduced the phishing detection rate to nearly zero. The study also proposed the concept of vulnerability level, which is defined as the ease with which phishing instances can evade the machine learning classifiers by manipulating features according to the cost and complexity of the manipulation. In addition, the researchers noted that many of the current methods for detecting phishing are primarily concerned with boosting the accuracy of the classification and neglect the robustness to adversarial attacks as an important aspect of contemporary AI-based phishing defenses [29].

iv. Human Factors and User Awareness in Phishing Attacks

Desolda et al. undertook a systematic literature review on the human factors involved in phishing attacks and the role of user behavior in the success of a cyber threat. The study showed that in addition to technical weakness, phishing attacks are frequently more effective due to other vulnerabilities, including lack of awareness, stress, distraction, trust, and limited knowledge of cybersecurity. The researchers reviewed various studies related to phishing susceptibility, organizational vulnerability, user awareness, and phishing defense methods. The findings showed that human errors contribute significantly to cybersecurity incidents, while social engineering techniques remain one of the most effective methods used in phishing attacks. The study also showed that user education, awareness training, and interactive learning approaches such as game-based training can improve phishing resistance and reduce the success of phishing attacks. Furthermore, the researchers suggested that future phishing defense systems should combine both technical detection mechanisms and human-centered security strategies to address technological and human-related vulnerabilities [30].



VIII. Real-World Impacts & Case Studies:

i. Real-World Case Studies: Famous phishing incidents

In [24], the phishing attacks and their influence are thoroughly examined, emphasizing various detection strategies along with security hurdles. The paper talks about phishing techniques evolved using social engineering and maliciously sophisticated mechanisms, making them difficult to detect using traditional systems. The research emphasizes the value of integrating different detection methods, like behavioral analysis and content analysis, to achieve greater accuracy. The study discusses phishing techniques and countermeasures, which provides a useful context at large for understanding such modern attack types. However, the article is mostly conceptual and does not provide a detailed implementation framework or experimental validation, which limits how useful it is for system development. This creates a gap for projects that require practical, implementable solutions for phishing detection.

ii. AI-Generated Phishing URLs and Real-Time Detection

New research features a real-time AI phishing URL detection system called PhishHaven that can detect both human-crafted and AI-generated phishing URLs. To enhance the performance of detecting phishing, the system employed a multi-threading method with an ensemble machine learning model and lexical feature analysis and HTML encoding. Unlike conventional phishing detection systems, PhishHaven was developed specifically to detect phishing URLs created using artificial intelligence models like DeepPhish that are capable of evading traditional security systems. The framework also added a URL Hit mechanism to identify tiny URLs and provided URL classification support for real-time URL classification via browser plug-in environment. The experimental results showed that the proposed system achieved a detection accuracy of approximately 98% and has better performance than some of the existing lexical-based phishing detection systems. The report also emphasized that phishing attacks using AI are one of the emerging cybersecurity threats, as they can closely mimic URLs and defeat traditional rule-based solutions [31].

IX. Future Trends in Phishing Detection:

i. Future Trends in Phishing Detection: AI-based attacks, adaptive security systems

[25] contains examples of genuine Phishing attempts and then elaborates on methods that the attacker has used to deceive victims. Basically, it highlights how attackers prey on trust and urgency, plus recognizable branding, to successfully engage in their attacks. It also outlines the few practical measures you can take to avert them—looking out for signed links, steering clear of dodgy attachments, and verifying who is reaching out to you. It is the effort to showcase real-world events where this resource excels; it gives context. But it is more on prevention and identifying than defining it in technical characteristics, or constructing a framework to banish the same. This limitation underlines the need to leverage user awareness augmented with in-and-out detection approaches to construct a strong phishing defence approach for practical implementations.



Gap Analysis:-

Reference	Aspect Covered	Key Idea	Identified Gap
[1]	Definition & Lifecycle	Phishing involves luring, interaction, exploitation stages	Lacks real-world validation of detection effectiveness
[2]	Evolution of Phishing	Shift from simple emails to AI-driven attacks	No analysis of AI impact on detection systems
[3]	Types of Attacks	Email, spear, whaling, clone phishing	No adaptive solutions for hybrid attacks
[6]	Email Spoofing	Header manipulation to impersonate senders	Limited evaluation of modern defenses
[7]	URL	Encoding, shortening, misleading URLs	Lacks real-time detection evaluation

Table I presents an overview of the fundamental aspects of phishing attacks, which include the definition, types, and techniques used in the process, such as email spoofing and URL manipulation. From the results obtained, phishing attacks heavily depend on social engineering tactics. The study reveals that most of the past literature focused on descriptive analysis without practical validation. Moreover, there was limited information regarding the application of AI in phishing detection.



Reference	Feature Type	Description	Limitation
[5]	Structural Features	Headers, body, URLs used for detection	Weak when used individually
[8]	Psychological Features	Urgency and fear tactics	Lack of countermeasure strategies
[9]	Linguistic Features	Tone, grammar, writing style	Ineffective against advanced phishing

Table II summarizes the main characteristics used in phishing email detection, including structural, psychological, and linguistic features. The findings indicate that combining multiple features improves detection accuracy. However, existing studies often evaluate these features independently and fail to address their robustness against sophisticated phishing emails.

Reference	Human Factor	Contribution	Gap
[4]	Social Engineering	Exploits user behavior and trust	Limited integration with technical detection
[22]	User Awareness and Training	Simulation-based learning improves awareness	Lacks integration with automated systems
[30]	Human Factors & User Awareness	User education and awareness training improve phishing resistance and reduce phishing success rates	Limited integration between human-centered security approaches and automated phishing detection systems

Table III highlights the role of human factors in phishing attacks, emphasizing social engineering and user awareness. The results show that phishing primarily targets human vulnerabilities. However, most research focuses on training users rather than integrating human behavior with intelligent detection systems.



Reference	Technique Type	Strength	Weakness
[11]	Rule-Based	Simple and low costs	Poor against new attacks.
[12]	Blacklist and Whitelist	Fast and easy implementation	Cannot detect new URLs
[13]	Machine Learning	Learns patterns from data	Depends on data quality
[14]	Deep learning	Detects complex patterns	Requires large data and resources.
[16]	NLP	Analyzes textual content	Weak against AI-generated emails
[26]	BERT Transformer Model	Strong contextual understanding and improved phishing classification accuracy	Requires advanced deep learning resources and large datasets
[27]	Transformer-Based Models (BERT, DistilBERT, RoBERTa)	Better performance than traditional ML models in understanding phishing email context	Traditional ML models struggle with evolving phishing language
[28]	Hybrid Deep Learning Framework	High detection accuracy and improved adaptability against evolving phishing attacks	Complex implementation and high computational requirements

Table IV presents different phishing detection techniques. While these methods improve detection, each has limitations such as low adaptability, data dependency, and high computational cost. No single method is sufficient alone.



Reference	Feature Category	Role	Gap
[15]	Feature Extraction	Combines text, URL, metadata features	May miss distributed attack signals.
[17]	Dataset Usage	Used for training models	outdated datasets.
[10]	Header Analysis	Detects spoofing via metadata	Underutilized despite high reliability.

Table V focuses on feature engineering and datasets. The findings show that combining multiple features improves performance. However, outdated datasets and underuse of header features limit effectiveness.

Reference	Challenge Type	Issue	Impact
[19]	System Limitations	High false positives	Reduces trust in detection systems
[20]	Zero-Day Attacks	Unknown attack patterns	Difficult to detect
[21]	Adversarial Attacks	ML model evasion	Reduces system accuracy
[29]	Evasion Attacks on ML-Based Phishing Detection	Feature manipulation can bypass machine learning classifiers	Significantly reduces phishing detection accuracy and system robustness
[31]	AI-Generated Phishing Threats	AI-generated phishing URLs can bypass traditional rule-based systems	Reduces the effectiveness of conventional phishing detection methods

Table VIII outlines major challenges in phishing detection. Existing systems struggle with evolving threats, zero-day attacks, and adversarial manipulation, highlighting the need for more adaptive solutions.



Study [18] discusses evaluation metrics in the detection of phishing including accuracy, precision, recall, and F1-score and they have been shown. It is clear that accuracy is not sufficient since the datasets used are imbalanced.

Paper [23] presents email security mechanisms and adaptive phishing detection using AI. According to the paper, it is possible to achieve better results when combining machine learning techniques and NLP technologies than applying other methods traditionally used for the same purpose.

Research [24] presents real-life examples of phishing attacks and their detection. This article talks about modern phishing tactics and recommends the use of more than one detection technique to achieve accuracy. Nonetheless, this work is rather conceptual and does not provide an actual phishing detection system.

Article [25] Identifies emerging trends in detecting phishing attacks, emphasizing AI-based attacks and adaptive security technologies. The research provides examples of phishing attacks and techniques to prevent them, but it emphasizes the need for awareness rather than detection systems.

Proposed Framework :-

I. Framework Overview

As one of the most pervasive and harmful forms of attacks, phishing emails have been a major challenge for automatic detection due to the limitations of conventional systems which rely on pre-defined rules or features. The proposed framework tackles the problem by integrating machine learning models with NLP techniques, URL features, and email headers in one solution. Instead of considering just one aspect of a phishing email, the proposed solution analyzes the content of an email from multiple perspectives simultaneously, taking into account linguistic characteristics, URLs embedded within the email, its sender, and urgency/urgency. These features make it possible to build a more robust classifier compared to other, more naive solutions. The ultimate objective is to train a Random Forest classifier to classify an email as phishing or not automatically. The proposed solution is meant to be practical and easy to reproduce, since it is completely built on top of Google Colab using only Python code.



II. System Architecture Diagram

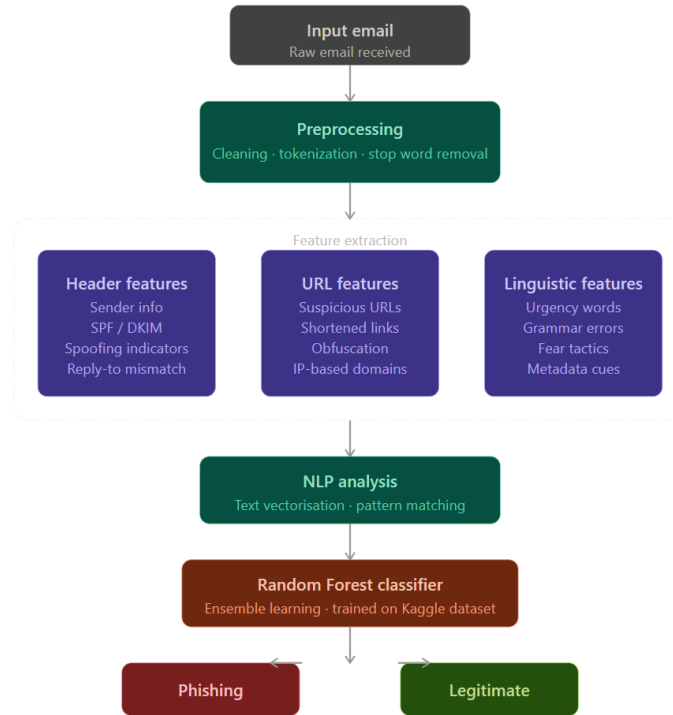


Figure 1. Proposed Phishing Email Detection Framework Architecture.

This model is designed based on a six-stage pipeline architecture. First, a raw email is taken in through the preprocessing stage. Here, the email is processed and converted into a form that is ready for further analysis. Then, the processed email is subjected to feature extraction where features are identified from three different perspectives: header-related features (sender information, SPF/DKIM), URL-related features (suspicious URLs, URL shortening), and linguistic features (urging language, spelling mistakes). All features identified are fed to the NLP analysis stage where textual emails are vectorized for processing by the classifier algorithm. Next, the output is generated by Random Forest as Phishing or Legitimate.



I. Preprocessing Phase

Prior to performing the analysis, all emails undergo a series of transformations to ensure consistency and usefulness of the data. The first transformation involves cleaning, during which duplicate entries are eliminated from the dataset, as well as those emails that lack important information. Afterward, links are extracted from the email message separately since URLs are analyzed separately as a distinct feature category. The email message itself is subsequently split into two main components, namely, header and message body, as the former and latter contain distinct signals and thus require distinct analysis techniques. The last step in the transformation process involves removal of all superfluous symbols such as punctuation marks and non-alphanumeric symbols from the email message, leaving only relevant textual information.

II. Feature Extraction Phase

The Feature Extraction Phase is viewed as one of the most significant steps within the suggested phishing email detection framework. The purpose of this stage is to extract a number of features related to phishing from emails, which includes content, URL, header, and other data available. Having extracted several types of features makes the task of distinguishing a phishing email from the regular one easier for the system.

i. Header-Based Features

Analysis based on header is all about looking into the technical attributes found in the header of an email. The following header-based attributes will be analyzed:

- Sender information analysis
- SPF validation (Sender Policy Framework)
- DKIM verification (DomainKeys Identified Mail)
- Email spoofing
- Mismatch between sender domains
- Abnormal "Reply-To"

The above list of attributes helps detect forged sender identification and spoofed email origins.

ii. URL-Based Features

Features based on the URL are applied in the analysis of the hyperlink embedded in emails. The use of fake or misleading URLs is common in the construction of phishing emails.



The URL-based features considered are as follows:

- Suspicious URLs
- Fake links
- URL manipulation tricks
- Multiple sub-domains
- Misspellings in domain name
- Misleading hyperlinks

Such analysis is applied to detect malicious links used in phishing emails.

iii. Linguistic Features

Language analysis concentrates on the detection of dubious linguistic patterns that are used in phishing emails. Hackers can manipulate the language in order to make the victim act immediately.

The linguistic features selected by extraction methods are:

- Emergency statements, e.g., “Act now” or “Immediately”
- Errors in grammar and spelling
- Threats and fear tactics
- Masking as legitimate companies
- Unusual ways of writing
- Punctuation and capitalization errors

This list comprises language elements that can be found in deceptive emails and help detect psychological manipulation.

iv. Metadata Features

Metadata analysis studies technical metadata associated with email transmission and communication patterns.

The list of metadata attributes comprises:

- Email timestamps
- Routing metadata
- Mismatch of reply-to header fields
- Metadata on sender IP addresses
- Patterns in email transmission



Metadata attributes assist in detecting abnormalities in communication patterns.

III. NLP Analysis Phase:

The NLP Analysis Phase will be geared towards analyzing the email text to find patterns of suspicious writing and phishing behaviors. NLP helps to improve phishing detection through analysis of the way the email message is written as opposed to only considering the technical aspects.

NLP analysis involves:

- Sentiment analysis
- Detection of suspicious writing
- Language pattern analysis
- Contextual text analysis
- Identifying manipulative language

The above-mentioned NLP analysis methods help the system detect any emails that employ persuasive language, emotional manipulation, or deception in communication.

IV. Classification Phase:

During the classification phase, the selected features are fed to train a machine learning classifier to identify phishing emails. The Random Forest algorithm is chosen for use because it is accurate, robust and handles multiple types of features simultaneously.

The following steps are carried out during the classification phase:

- Training the Random Forest classifier using the data
- Learning phishing patterns from extracted features
- Email classification into pre-defined categories

Emails are classified into:

- Phishing Email
- Legitimate Email

V. Output Phase:

The output phase is the last step in our proposed framework for detecting phishing emails. The final prediction for each email is obtained by this framework after classification.

Output will be in the form of:

- Phishing Warning
- Legitimate Email Alert



If any features of phishing are identified, then phishing warnings will be provided by the system.

If no such features exist, then the email will be classified as legitimate.

The other possible outputs might be:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The above outputs can be used to evaluate the performance of the proposed framework.

Methodology :-

I. Research Design

This research follows a quantitative experimental approach for detecting phishing emails using machine learning techniques. The study focuses on identifying common phishing characteristics such as suspicious URLs, urgency language, spoofed sender addresses, grammatical errors, and misleading attachments. A phishing email classification framework is proposed to distinguish phishing emails from legitimate emails based on extracted features.

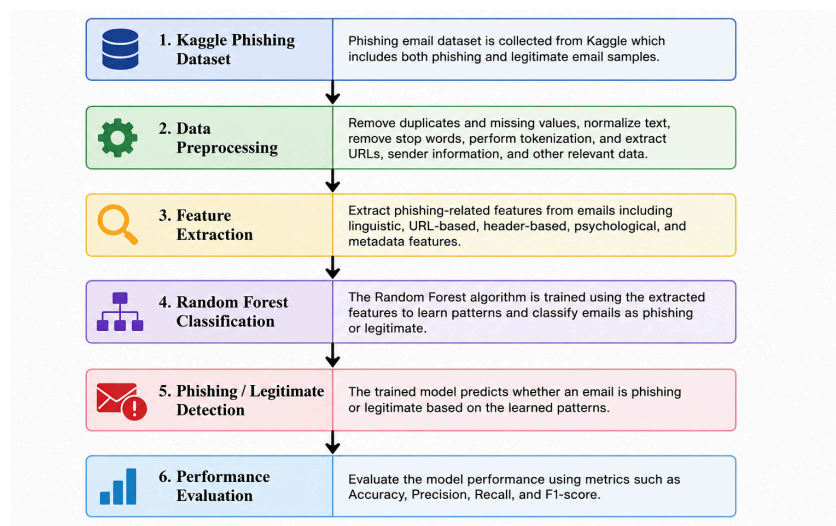


Figure 2. Proposed Methodology Workflow for Phishing Email Detection System



II. Dataset Collection

Phishing email corpus was downloaded from a publicly available Kaggle repository collecting phishing and legitimate emails from publicly available email accounts. The dataset contains several features that can be utilized for phishing email detection. These features include email text, sender information, URL, text, and subject of the email.

The dataset can be used to detect phishing, as it contains instances of various types of phishers and legitimate emails which the model can use to distinguish the email content between phishing and non-phishing ones.

III. Data Preprocessing

Before the classification model is trained, the dataset undergoes various preprocessing steps. These steps generally include the elimination of duplicate records and records with missing values. Some steps, such as lowercasing, stripping punctuation symbols, and removing other non-alphanumeric characters from the email text, can be termed text normalization processes.

URLs, sender addresses and suspicious keywords are extracted from emails. Stop words are removed from the email, since they account for most words in every email and thus add no meaningful value. Tokenization and other preprocessing techniques are used to improve feature extraction and machine learning performance.

IV. Feature Extraction

Feature extraction is one of the most important stages in phishing email detection because phishing attacks often contain multiple suspicious indicators distributed throughout the email structure. This study extracts several categories of phishing-related features based on findings from previous literature.

The extracted features include:

- Linguistic cues include errors of grammar, unusual writing styles, urgent words, and threatening words.
- Other URL-based features include shortened URLs, suspicious domains, excessive subdomains, and misleading hyperlink text.
- Header-based features such as spoofed sender addresses, mismatched domains, and abnormal reply-to addresses.
- Psychological indicators involving fear tactics and urgency expressions like “act now” or “immediately.”
- Metadata features including sender information and routing-related attributes.



Combining multiple feature categories improves phishing detection accuracy and reduces dependence on a single indicator.

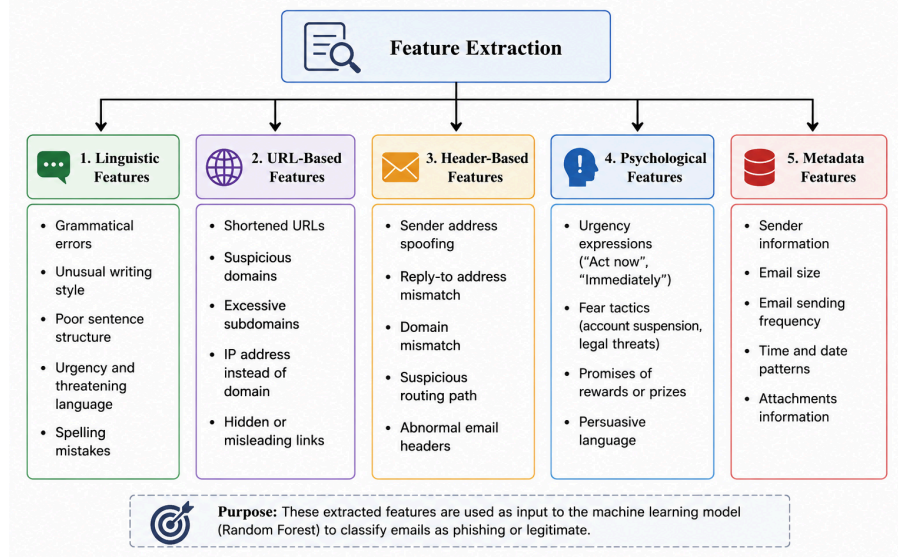


Figure 3. Feature Extraction Categories Used in the Proposed Phishing Detection Framework

V. Proposed Detection Model

The authors propose a novel framework for phishing email detection. The Random Forest machine learning algorithm was chosen for the project, due to its high classification accuracy, ability to handle several different features, and ability to reduce overfitting via ensemble learning.

The proposed framework consists of two categories of emails:

- Phishing email
- Legitimate email

The proposed model utilizes the features extracted from email contents, URLs and email metadata in order to detect the suspicious behavioral patterns of emails. The model is expected to be simple and efficient, but accurate.

VI. Experimental Environment

The proposed framework is intended to be implemented using Google Colab as the experimental environment. Python programming language is used together with machine learning and data analysis libraries such as:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib



Google Colab provides a cloud-based environment that supports machine learning experiments without requiring high-end hardware resources.

VII. Evaluation Metrics

The performance of the proposed phishing detection framework is evaluated using several standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score

Accuracy is the proportion of the emails classified correctly by the classifier. Precision is the proportion of predicted phishing emails that were actual phishing emails. Recall is the system's ability to correctly detect phishing emails. The F1-score is the harmonic mean of the precision and recall.

Moreover, accuracy is often insufficient for phishing datasets, as there is a possibility of imbalances between the class distributions.

VIII. Experimental Procedure

The proposed phishing detection framework contains multiple stages:

1. Download the phishing email dataset from Kaggle.
2. Clean and normalize email data for further analysis.
3. Extract phishing features from email content, URLs, and metadata.
4. Train a Random Forest classifier on the processed dataset.
5. Classify emails as phishing or legitimate.
6. Evaluate model performance using standard evaluation metrics.

This workflow can be used to detect phishing emails based on multiple phishing features and classification methods.

Expected output:-

The expected output of the proposed phishing email detection framework is a trained machine learning model capable of classifying emails into two categories: phishing emails and legitimate emails. After preprocessing the Kaggle phishing email dataset and extracting phishing-related features, the Random Forest classifier is expected to analyze email characteristics and produce a final prediction for each email.

The system output will include the predicted email class, either “Phishing” or “Legitimate.” Accuracy, precision, recall, F1-score, and a confusion matrix are further predicted to be used to assess the model's performance. These outputs help measure how effectively the proposed framework can detect phishing emails and reduce incorrect classification.



The expected results should show that using multiple feature categories, such as linguistic features, URL-based features, header-based features, psychological indicators, and metadata features, improves the ability of the model to identify phishing emails compared to relying on one feature type only.

Expected Output	Description
Email Classification	Predicts whether an email is phishing or legitimate
Accuracy	Measures the overall correctness of the phishing detection model
Precision	Measures how many emails predicted as phishing are actually phishing
Recall	Measures how many actual phishing emails are correctly detected
F1-score	Provides a balance between precision and recall
Confusion Matrix	Displays correct and incorrect email classifications performed by the model

Table 1. Expected Outputs of the Proposed Phishing Email Detection Framework

Dataset Preview:

	sender	receiver	date	subject	body	label	urls
0	Young Esposito <Young@lworld.de>	user4@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 16:31:02 -0700	Never agree to be a loser	Buck up, your troubles caused by small dimensi...	1	1
1	Mok <ipline's1983@icable.ph>	user2.2@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 18:31:03 -0500	Befriend Jenna Jameson	InUpgrade your sex and pleasures with these te...	1	1
2	Daily Top 10 <Karmandeep- opengevi@universalnet...	user2.9@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 20:28:00 -1200	CNN.com Daily Top 10	>+#+#+#+#+#+#+#+#+#+#+#+#+#+#+#+#+#+#+#+...	1	1
3	Michael Parker <lvqmail@pobox.com>	SpamAssassin Dev <xrh@spamassassin.apache.org>	Tue, 05 Aug 2008 17:31:20 -0600	Re: svn commit: r619753 - in /spamassassin/tru...	Would anyone object to removing .so from this ...	0	1
4	Gretchen Suggs <externalsep1@loanofficertool.com>	user2.2@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 19:31:21 -0400	SpecialPricesPharmMoreinfo	InWelcomeFastShippingCustomerSupportInhttp://7...	1	1

Figure 4. Dataset Preview in Google Colab



Model Performance:

Accuracy: 0.9962

Precision: 0.9968

Recall: 0.9963

F1-score: 0.9966

Classification Report:

	precision	recall	f1-score	support
Legitimate	1.00	1.00	1.00	3462
Phishing	1.00	1.00	1.00	4369
accuracy			1.00	7831
macro avg	1.00	1.00	1.00	7831
weighted avg	1.00	1.00	1.00	7831

Figure 5. Random Forest Model Performance Metrics

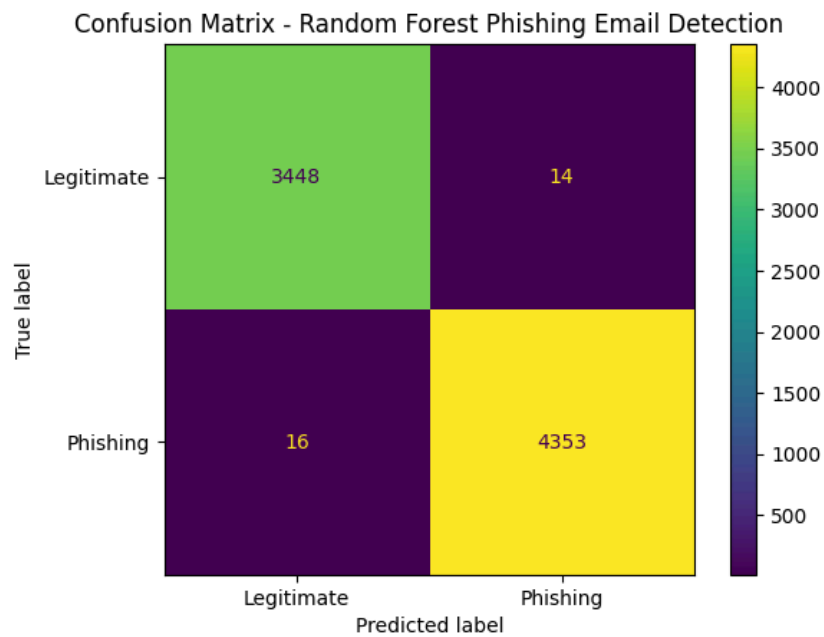


Figure 6. Confusion Matrix for Phishing Email Classification



Conclusion:-

To summarize, phishing emails still pose one of the gravest cyber security issues as they take advantage of both technical vulnerabilities and human actions. The project investigated the typical features of phishing emails, such as questionable URLs, fake from-email addresses, creating urgency and fear, bad grammar and spelling mistakes, misleading attachments, and malicious Email header. The literature review showed how phishing attacks have evolved from simple email scams to well-organized, AI and social engineering-based attackers. Consequently, conventional detection methods such as blacklists and rule-based systems alone aren't adequate.

The proposed phishing detection framework uses the machine learning approach, specifically Random Forest classifier, to examine several phishing features at the same time. As mentioned above, the framework utilizes linguistic, URL-based, header-based, psychological and metadata features to enhance phishing detection accuracy while reducing false positive cases. The paper emphasized the requirement of preprocessing, feature extraction and evaluation matrix (Accuracy, precision, recall and F1-score) to form an efficient phishing detection system.

The project also stressed the fact that technical solutions alone are not enough for phishing prevention. Training users, providing cybersecurity education, and design strategies that are centered on the human element still matter as attackers often use trusted sources or urgency to invade; they rely heavily on emotional manipulation. This research argued that the discovery of clever phishing attacks e.g. phishing URLs generated by machine learning, adversarial (ML), ML methods to deceive & mislead human followers are growing challenges for cybersecurity researchers and organizations.

In summary, the project shows that a promising way to detect phishing emails is to apply machine learning techniques, where multi-feature analysis from has so far provided reasonable results. Future research could explore real-time detection implementations, implementation with an improved deep learning and transformer-based models, it would also focus on zero day phishing attacks where no prior dataset is available, availability of intelligent and exogenous surrounding information for real-time integration along with user awareness training to create more robust in terms of adaptive cybersecurity system.



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